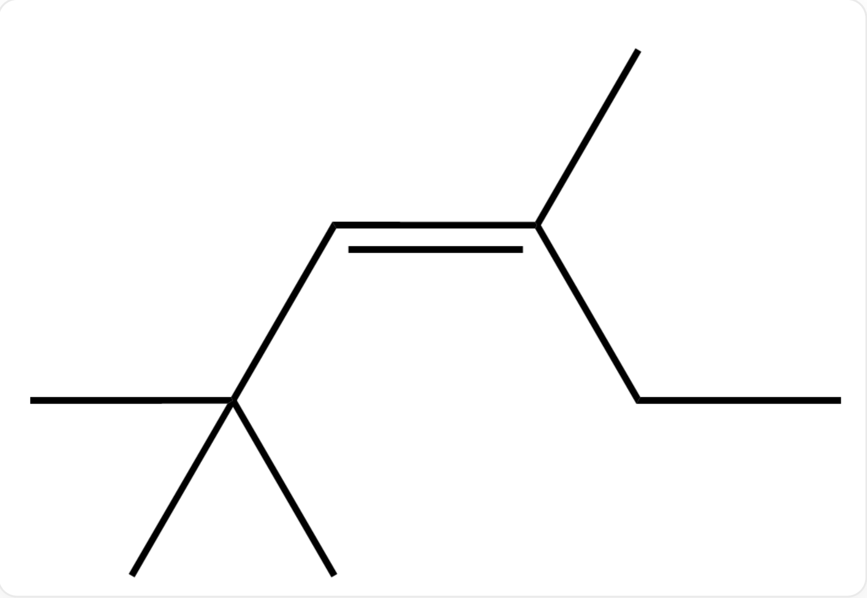
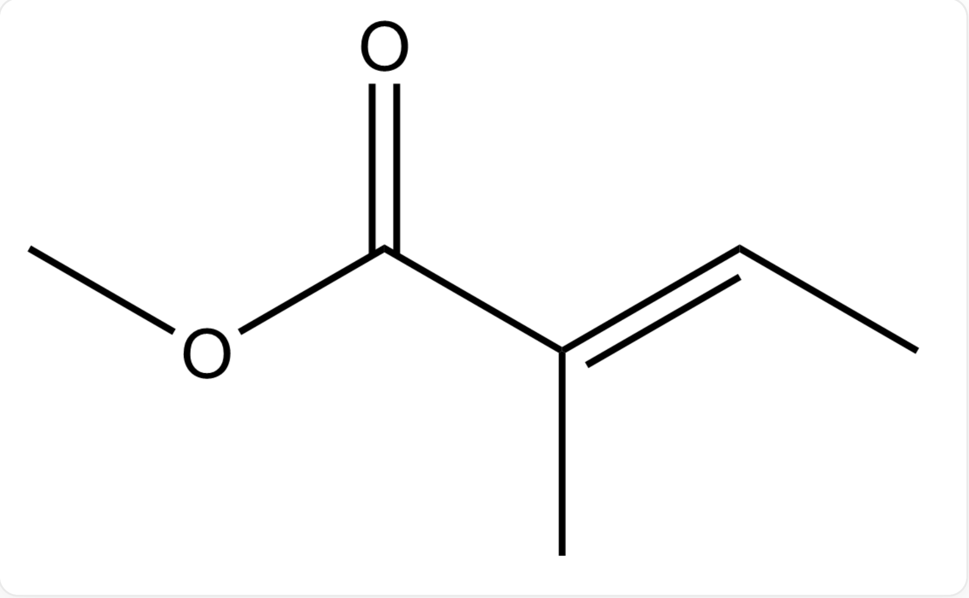
题目

忽略立体化学，请分别预测单线态氧 $^1O_2$ 与以下两种底物反应的主产物



C/C(CC)=C/C(C)(C)C,底物1

底物1

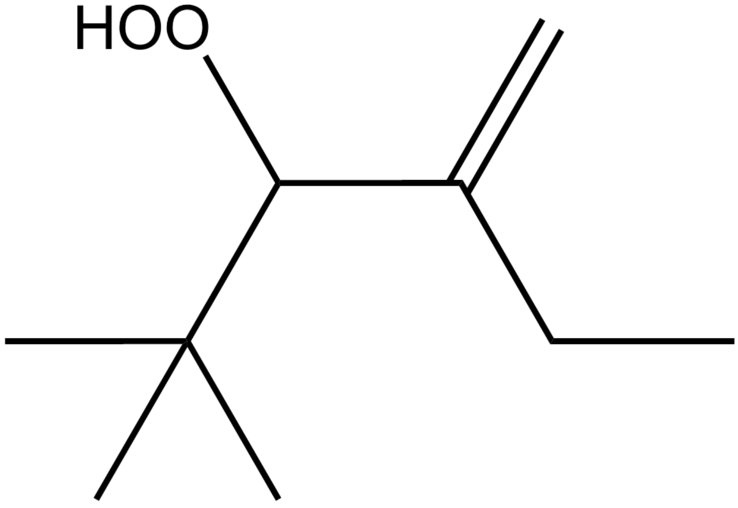


O=C(/C(C)=C/C)OC,底物2

底物2

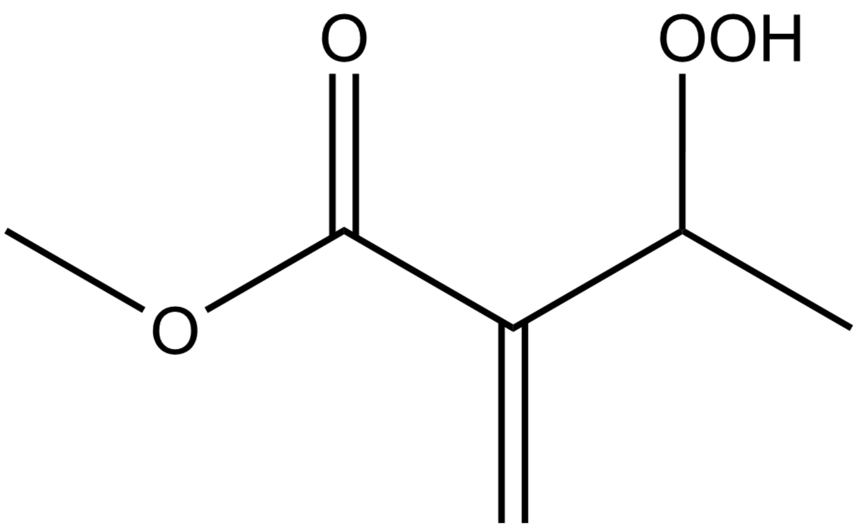
A. 其他选项均不正确

B.



CC(C)(C)C(OO)C(CC)=C,产物1

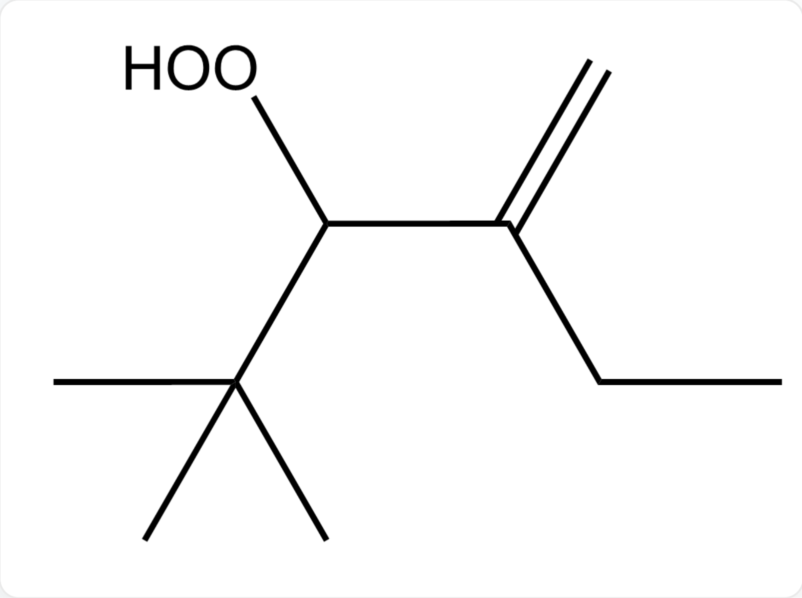
产物1



O=C(C(C(OO)C)=C)OC,产物2

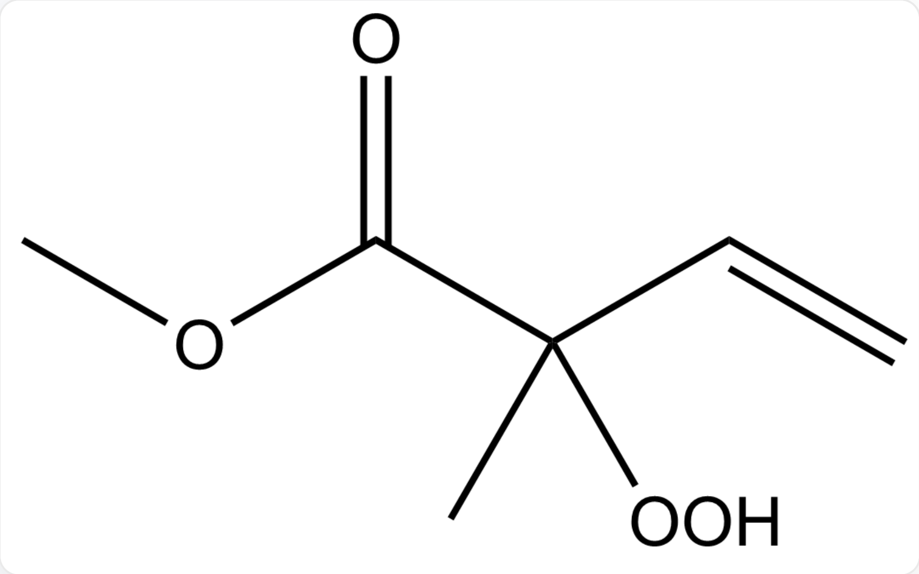
产物2

C.



CC(C)(C)C(OO)C(CC)=C,产物1

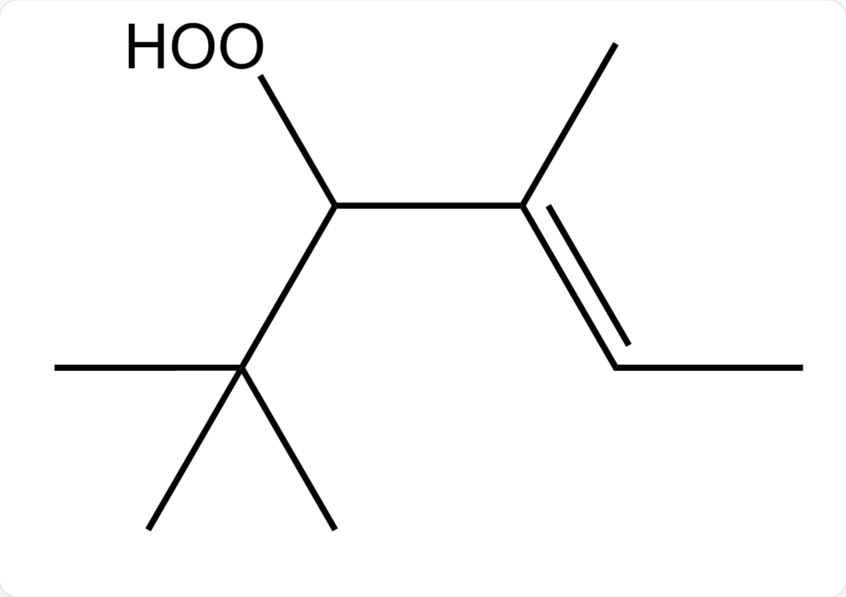
产物1



O=C(C(C)(OO)C=C)OC,产物2

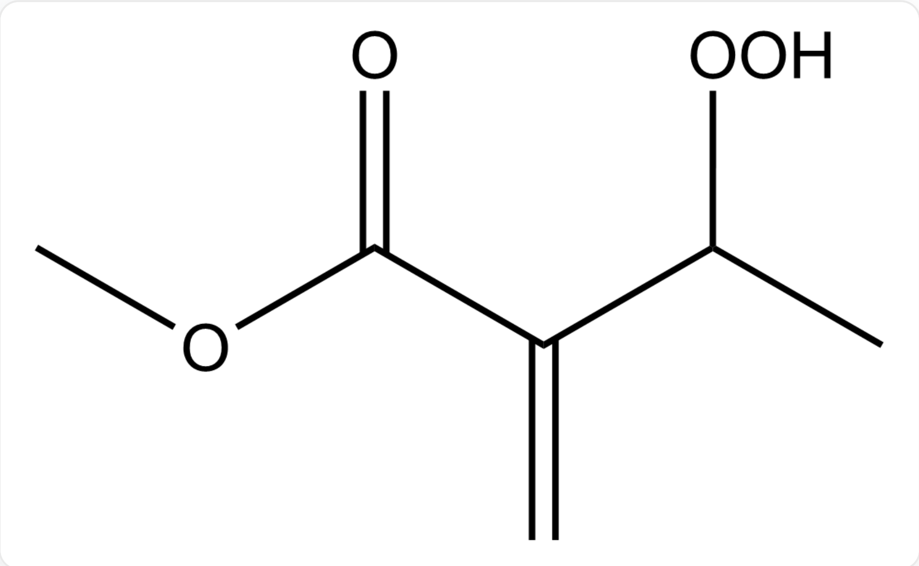
产物2

D.



CC(C)(C)C(OO)/C(C)=C/C,产物1

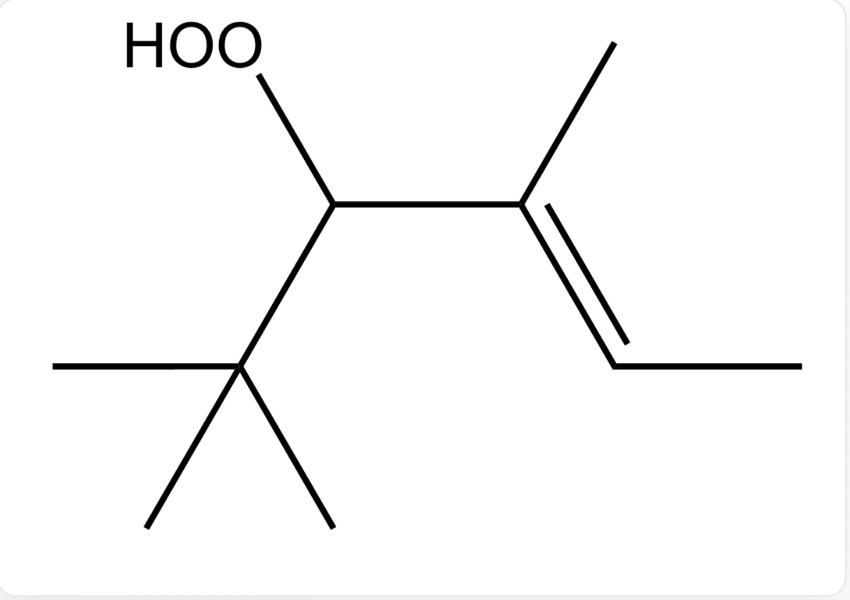
产物1



O=C(C(C(OO)C)=C)OC,产物2

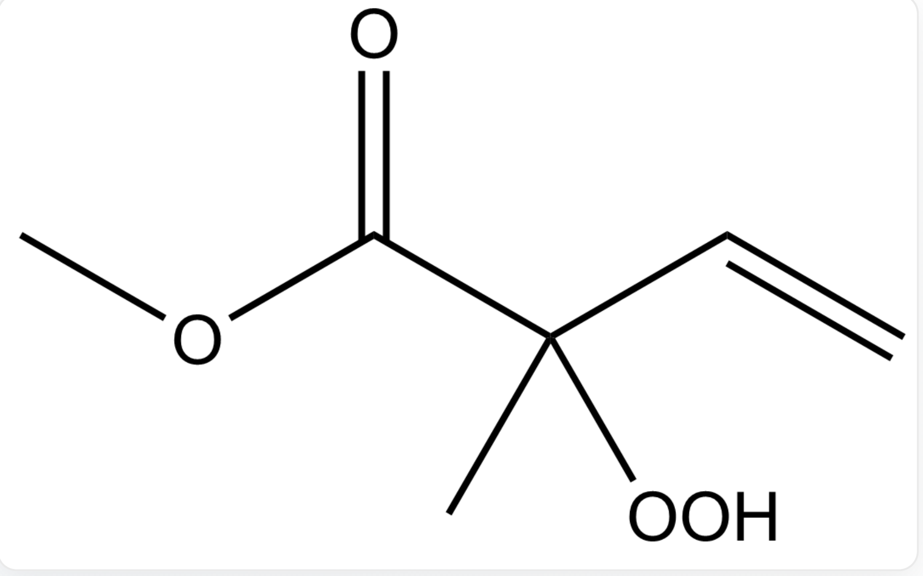
产物2

E.



CC(C)(C)C(OO)/C(C)=C/C,产物1

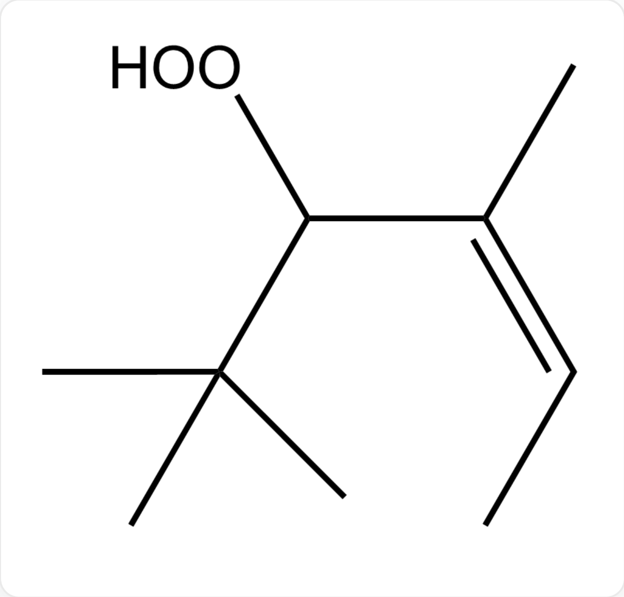
产物1



O=C(C(C)(OO)C=C)OC,产物2

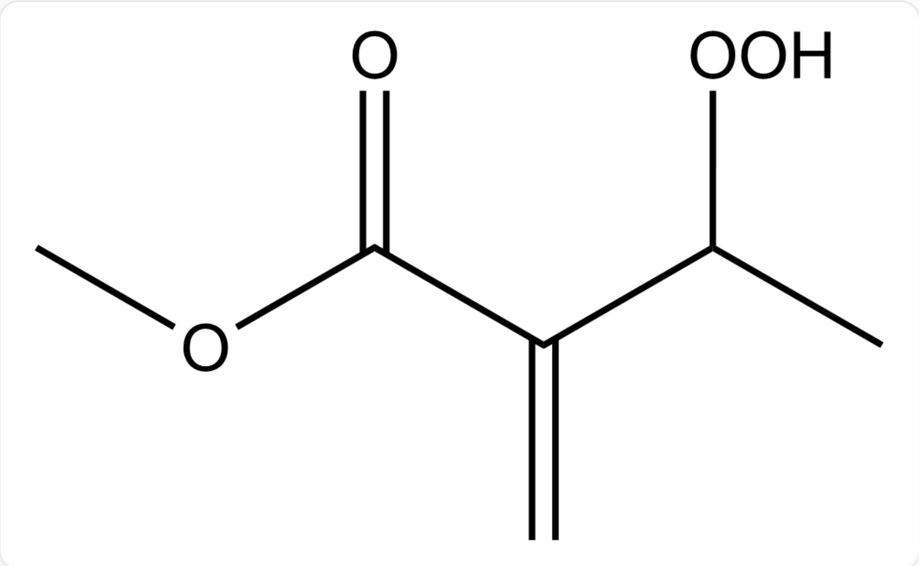
产物2

F.



CC(C)(C)C(OO)/C(C)=C\C,产物1

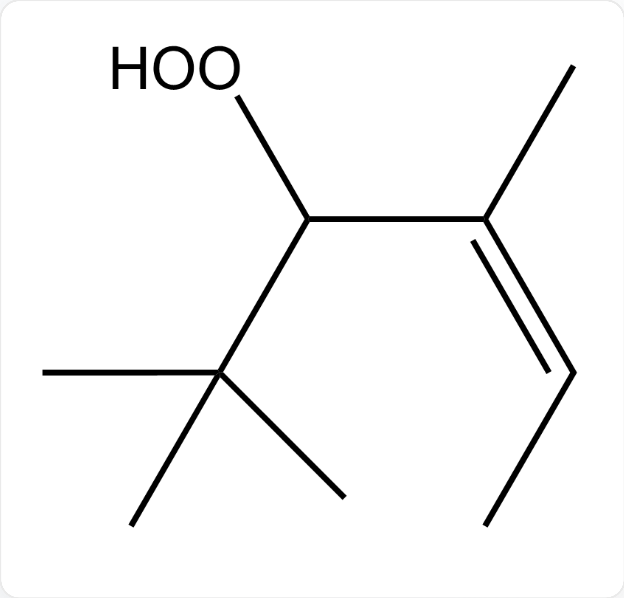
产物1



O=C(C(C(OO)C)=C)OC,产物2

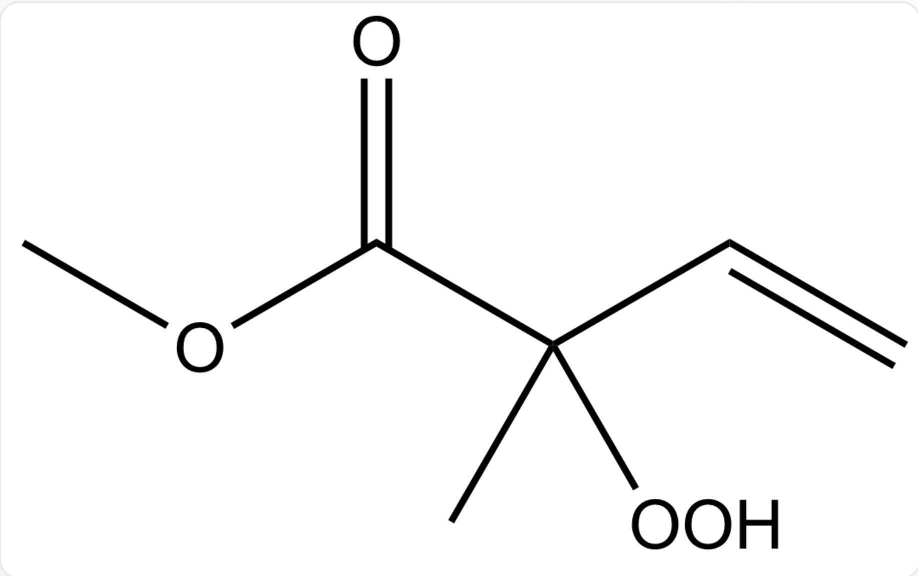
产物2

G.



CC(C)(C)C(OO)/C(C)=C\C,产物1

产物1



O=C(C(C)(OO)C=C)OC,产物2

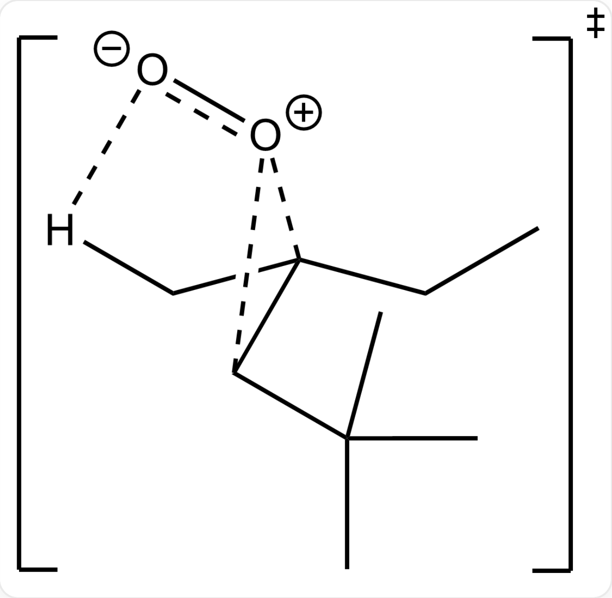
产物2

答案

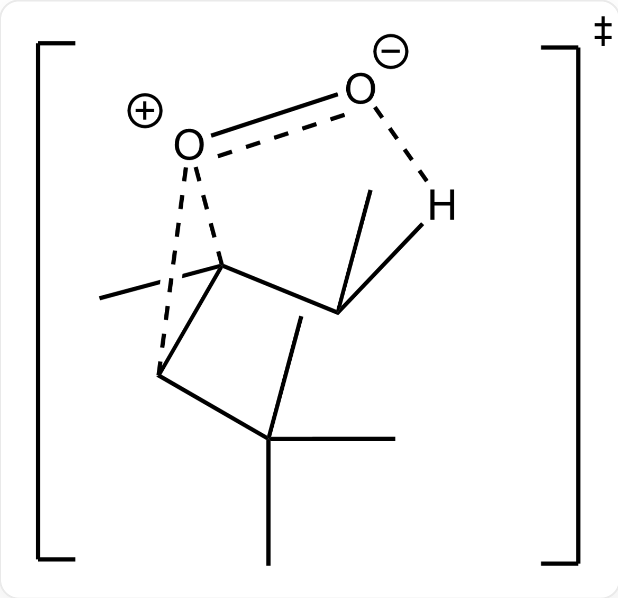
正确答案: B

详细解析

对于反应1来说，两种过渡态如下图所示



[O-][O+]1C(C(C)(C)C)C1(C[H])CC



[O-][O+]1C(C(C)(C)C)C1(C[H])CC

两种过渡态中均只存在一组$O-H$相互作用

CHECKPOINT 1 PTS

两种过渡态中均只存在一组$O-H$相互作用

但是叔丁基朝上的甲基会与氧发生排斥，进一步提高过渡态能量

CHECKPOINT 1 PTS

但是叔丁基朝上的甲基会与氧发生排斥

对于反应2

进攻羰基邻位的氢,羰基可通过共轭效应稳定过渡态,故得到共轭产物

CHECKPOINT 1 PTS

进攻羰基邻位的氢,羰基可通过共轭效应稳定过渡态,故得到共轭产物